

# Fold a Fitted Sheet SOP (1x Episode: 1 Fitted Sheet)

One episode folds one fitted sheet. The sheet starts spread flat on the fold mat with its four corner pockets facing up. The four pockets are nested into one stack, the loose elastic hem is tucked in, the sheet is stroked flat, folded in thirds one way and then in thirds the other way, and the finished packet is squared up.

The order never changes: nest, tuck, flatten, thirds across, thirds front to back, square. Each step is finished and checked before the next one starts.

The sheet never leaves the fold mat. Nothing is folded in the air, nothing is pushed or shoved into place, and a gripper opens only over the mat.

## Setup

Complete both checklists before starting an episode.

### Cell configuration

- **Environment camera:** 900 mm.
- **cell\_type:** bimanual

### Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the fold mat at the front center with the sheet spread on it, and bare tabletop all round.
3. The overhead camera shows the whole fold mat, so every corner, every hem, and every fold edge can be seen landing.
4. All four corners of the spread sheet are in frame at the same time.
5. Both arms are at home with grippers open.
6. The tabletop holds nothing except the fold mat, the sheet, and robot hardware.
7. The right arm reaches all four corners of the spread sheet, the whole fold mat, and both hem sides. It does not lean out or stretch to a joint limit to get there.
8. The left arm reaches all four corners of the spread sheet and the whole fold mat. It does not lean out or stretch to a joint limit to get there.

### Materials checklist

1. One small fitted sheet, cot or crib size, so it lies spread inside the fold mat with clear mat all round.

2. The sheet lies **pockets up**, spread flat and square to the mat, with no corner folded under and no part of it off a mat edge.
3. All four **corner pockets** are open and hold their shape. Each mouth stands open when its hem is lifted, instead of staying shut.
4. The **elastic hem** runs all the way round the sheet, unbroken, with no loose thread and no place where the elastic has pulled out of the seam.
5. The sheet is dry, clean, and pressed flat, with no tears and no open seams. It holds a fold instead of springing open, and it shows no fold line from an earlier episode.
6. The **fold mat** lies flat at the front center of the table. Its surface grips cloth, so the sheet does not slide when a gripper strokes it. It does not shift when a gripper presses and drags.
7. Nothing else is on the mat.

## Workspace layout

- **Fold mat:** flat at the front center of the table, where both arms reach the whole mat. Every nest, tuck, stroke, and fold happens here. It holds one sheet at a time. The finished packet is left on it.
- **Mat sides:** the mat's four sides are named **far**, **near**, **left**, and **right**. The near side is the one closest to the front edge of the table. Every direction in this document is named from these four sides.
- The rest of the tabletop stays bare for the whole episode. Nothing is staged on it and nothing is set down on it.

The mat position stays the same for every episode.

## Arm assignments

- **Right gripper:** works. It carries every travelling corner into its pocket, tucks the far hem, strokes the sheet flat, folds the right third and both front-to-back thirds, and squares the finished packet.
- **Left gripper:** supports. It holds each receiving pocket open, pins the sheet for every fold and every tuck, and folds and strokes on the left side only: the left hem tuck and the left third.
- Wherever the left gripper holds, the right gripper works clear of it. The two grippers never crowd the same spot.
- Neither gripper reaches past the far side of the mat, and neither one lifts or moves the mat.

## Vocabulary

- **Pockets up:** the sheet lies with its four corner pockets facing the ceiling, so every pocket mouth can be seen and reached from above. The sheet is staged this way and stays this way.

- **Corner pocket:** the sewn corner of a fitted sheet. It is a flap of cloth with an opening, and the opening faces the middle of the sheet when the sheet is spread pockets up.
- **Pocket mouth:** the opening of a corner pocket. It is the only way into the pocket.
- **Seam point:** the sewn tip of a corner, where the two side seams meet. It is the only part of a corner a gripper pinches when carrying it.
- **Elastic hem:** the gathered band running all the way round the edge of the sheet. It bunches into a wavy edge when the sheet lies flat.
- **Gape:** the left gripper pinches a pocket's hem, lifts it up and back, and holds it there, so the pocket mouth stands open. A gape is held for the whole substep. It is not a pull and let go.
- **Nest:** sliding one corner all the way inside another corner's pocket, so the outer pocket covers the corner inside it and the two seam points sit together.
- **Nest stack:** the corners that have already been nested into each other, treated from then on as one corner. It is picked up at its outer seam point, like any other corner.
- **Pin:** the gripper closes, comes straight down, and presses on the cloth to hold it still. It stays there without moving until the step says it can lift. A pin does not squeeze the cloth between the fingers and it does not slide along.
- **Stroke:** the gripper closes, presses down, and runs along in one smooth line from one end to the other, staying in contact the whole way. It never lifts off in the middle.
- **Fold:** a gripper pinches an edge or a corner, lifts it clear of the cloth under it, carries it across, lowers it onto the spot it is going, and opens. Nothing is ever folded by pushing it along the mat.
- **Fold edge:** the closed crease a fold makes. The other sides of a folded panel are loose open edges.
- **Lift clear:** raise the cloth straight up until there is daylight under it, before anything moves sideways.
- **Bite:** one pinch, lift, fold, and lower, worked at one spot along a side. A tuck is made in three bites rather than one long grab.
- **Edge tuck:** folding the elastic hem inward onto the sheet, in three bites along a side, so that side of the sheet runs straight instead of wavy.
- **Outline:** the shape the sheet reads as from above. It runs straight once the tucks are done, and no hem bulges outside it.
- **Thirds:** three panels that look about the same width as each other, judged by eye. The first fold lands its edge about two-thirds of the way across. The second fold lands its edge level with the first fold edge.
- **Panel:** one of the three bands a thirds fold makes.
- **Packet:** the sheet once both thirds folds are done. It is one squared block of layers.
- **Resting:** the cloth is down and takes its own weight. It stays put when the gripper lifts clear, with no layer springing open.

- **Release point:** over the fold mat, once the cloth is already resting. A gripper opens nowhere else.

## Steps

Run Steps 1 to 8 once, in order. One sheet is on the mat for the whole episode and it never leaves the mat.

### Step 1: Check and straighten the spread sheet

**Goal:** the sheet lies spread flat and square on the mat, pockets up, with all four corners showing.

- Look at the sheet on the mat. It lies pockets up, square to the mat, with all four corner pockets showing and nothing off a mat edge.
- If a corner is folded under itself, the gripper on that side pinches it at its **seam point**, lifts it clear, carries it out, and lays it flat.
- If the sheet sits crooked on the mat, the **right gripper** pinches the nearest corner at its seam point, lifts it clear, and lays it down so the sheet sits square to the mat again.
- Both grippers lift away.

Straighten by lifting and laying down. Never drag the sheet across the mat to square it up.

**Check:** the sheet lies flat and square on the mat, pockets up. All four corner pockets are showing and open. No corner is folded under. Nothing is off a mat edge.

If the sheet is bunched or lying pockets down, stop the episode and stage it again from the reset list.

### Step 2: Nest the four corner pockets

**Goal:** all four corners nested into one **nest stack** at the far-left, with the sheet folded in quarters.

Run 2.1, 2.2, and 2.3 in that order. Each one uses the same pattern: the **left gripper** gapes the receiving pocket and holds it, and the **right gripper** carries the travelling corner in.

#### 2.1 Nest the right pair

- The **left gripper** pinches the **far-right pocket** at its hem, lifts it up and back, and holds that **gape**, so the pocket mouth stands open facing the front edge.
- The **right gripper** pinches the **near-right corner** at its **seam point**, lifts it clear of the mat, and carries it level to the far-right corner.
- The **right gripper** slides that corner into the open pocket mouth until its seam point sits deep inside, then opens.
- The **left gripper** lowers the outer pocket over the corner inside it and lifts away.

- The **right gripper** strokes once along the outer pocket's hem, far end to near end, so the outer pocket lies square over the corner inside it, then lifts away.

## 2.2 Nest the left pair

- The **left gripper** gapes the **far-left pocket** the same way and holds it.
- The **right gripper** pinches the **near-left corner** at its seam point, lifts it clear, carries it level to the far-left corner, and slides it into the open pocket mouth, then opens.
- The **left gripper** lowers the outer pocket over the corner inside it and lifts away.
- The **right gripper** strokes once along the outer pocket's hem, far end to near end, then lifts away.

The sheet now lies folded in half, with the near edge on the far edge and a fold edge along the near side.

## 2.3 Join the two stacks

- The **left gripper** gapes the outermost pocket of the **far-left nest stack**, so its mouth stands open facing right, and holds it.
- The **right gripper** pinches the **far-right nest stack** at its outer seam point, lifts it clear, carries it level to the left, and slides it into the open pocket mouth, then opens.
- The **left gripper** lowers the outer pocket over the stack inside it and lifts away.
- The **right gripper** strokes once along the outer pocket's hem, far end to near end, then lifts away.

Carry the stack level and let the sheet fold itself in half as the stack travels. Never drag the stack across the mat.

**Check:** there is one **nest stack** at the far-left, holding all four corners. No corner sits on the outside of a pocket. The sheet reads as a rectangle folded in quarters. The near side and the right side are fold edges. The far side and the left side carry the elastic hem.

If a corner sits on the outside instead of inside, the **left gripper** gapes that pocket again and the **right gripper** slides the corner in.

## Step 3: Tuck the elastic hem in

**Goal:** the far side and the left side run straight, with no hem bulging outside the **outline**.

Only the far side and the left side carry hem. The near side and the right side are fold edges and are never tucked.

## 3.1 Tuck the far side

- The **left gripper** closes and **pins** the sheet at its middle. It holds there for the whole substep.

- The **right gripper** works three **bites** along the far side, in this order: the left end, the middle, the right end. Each bite: pinch the hem, lift it clear, fold it inward onto the sheet so the far side runs straight, lower it, and open.
- The **right gripper** strokes along the new far side, left end to right end.
- Both grippers lift away.

### 3.2 Tuck the left side

- The **right gripper** closes and **pins** the sheet at its middle. It holds there for the whole substep.
- The **left gripper** works three bites along the left side, in this order: the far end, the middle, the near end. Each bite is the same: pinch the hem, lift it clear, fold it inward onto the sheet, lower it, and open.
- The **left gripper** strokes along the new left side, far end to near end.
- Both grippers lift away.

Fold the hem inward, onto the sheet. A hem folded outward or left hanging past the side puts the wave back in the outline.

**Check:** the far side and the left side run straight. No hem bulges outside the outline and no hem hangs off a side. The nest stack still sits at the far-left corner and no corner has come out of it.

If a stretch of hem springs back out, the gripper for that side takes one more bite at that spot and strokes the side again.

## Step 4: Stroke the sheet flat

**Goal:** the quartered sheet lies flat and even, with no bubble or ridge standing up.

### 4.1 Stroke out to the far, right, and near sides

- The **left gripper** closes and **pins** the sheet at its middle. It holds there for the whole substep.
- The **right gripper** strokes from beside the pin out to the **far side**, lifts off, comes back, strokes out to the **right side**, lifts off, comes back, and strokes out to the **near side**.
- Every stroke runs from the middle outward. Never stroke back toward the middle.

### 4.2 Stroke out to the left side

- The **right gripper** closes and **pins** the sheet at its middle. It holds there for the whole substep.
- The **left gripper** strokes from beside the pin out to the **left side**.
- Both grippers lift away.

**Check:** the sheet lies flat with no bubble, ridge, or standing wrinkle. All four sides run straight. The nest stack lies flat at the far-left. Nothing has been pushed off square: the near side and the right side still run straight and meet at about a right angle.

If the sheet has been shoved out of square, the **right gripper** pinches the corner that moved, lifts it clear, lays it back in line, and strokes that side again.

## **Step 5: Fold in thirds, left to right**

**Goal:** three panels of about the same width, one on top of the other, with the nest stack inside.

### **5.1 Fold the left third in**

- The **right gripper** closes and **pins** the sheet at its middle. It holds there for the whole substep.
- The **left gripper** pinches the **left side** halfway along it, lifts it clear, carries it right, and lowers it so the left side lands about **two-thirds of the way across** and the new fold edge runs straight from the far side to the near side. It opens once the panel is resting.
- The **left gripper** strokes along the new fold edge, far end to near end.
- Both grippers lift away.

The left third goes first, so the nest stack ends up in the middle of the packet instead of on the outside.

### **5.2 Fold the right third over**

- The **left gripper** closes and **pins** the sheet at its middle. It holds there for the whole substep.
- The **right gripper** pinches the **right side** halfway along it, lifts it clear, carries it left, and lowers it so the right side lands **level with the fold edge made in 5.1** and the new fold edge runs straight from the far side to the near side. It opens once the panel is resting.
- The **right gripper** strokes along the new fold edge, far end to near end.
- Both grippers lift away.

**Check:** the sheet is now about a third as wide as it was. The three panels look about the same width. Both fold edges run straight and parallel to each other. No panel hangs out past the layer under it. The nest stack is covered.

If a panel landed short, long, or crooked, the gripper that made that fold lifts that side, lays it down again, and strokes the fold edge.

## **Step 6: Fold in thirds, front to back**

**Goal:** the folded strip brought down to a packet of three panels, with the nest stack inside.

## 6.1 Fold the far third down

- The **left gripper** closes and **pins** the strip at its middle. It holds there for the whole substep.
- The **right gripper** pinches the **far side** halfway along it, lifts it clear, carries it toward the front edge, and lowers it so the far side lands about **two-thirds of the way toward the front edge** and the new fold edge runs straight from left to right. It opens once the panel is resting.
- The **right gripper** strokes along the new fold edge, left end to right end.
- Both grippers lift away.

The far third goes first, for the same reason: it puts the nest stack inside the packet.

## 6.2 Fold the near third over

- The **left gripper** closes and **pins** the strip at its middle. It holds there for the whole substep.
- The **right gripper** pinches the **near side** halfway along it, lifts it clear, carries it away from the front edge, and lowers it so the near side lands **level with the fold edge made in 6.1** and the new fold edge runs straight from left to right. It opens once the panel is resting.
- The **right gripper** strokes along the new fold edge, left end to right end.
- Both grippers lift away.

**Check:** the packet is now about a third as deep as the strip was. The three panels look about the same depth. Both fold edges run straight and parallel to each other. No panel hangs out past the layer under it.

If a panel landed short, long, or crooked, the **right gripper** lifts that side, lays it down again, and strokes the fold edge.

## Step 7: Square the packet

**Goal:** the packet reads as a rectangle with four straight sides and no layer sticking out.

- The **left gripper** closes and **pins** the packet at its middle. It holds there for the whole step.
- Work the four sides in this order: **far, right, near, left**. On each side, if a layer stands out past the layer under it, the **right gripper** pinches that layer, lifts it clear, and lays it back level with the layer under it.
- The **right gripper** then strokes along each of the four sides in the same order, one stroke per side, end to end.
- Both grippers lift away.

A layer that is out is lifted and laid down again. Nothing is patted, nudged, or shoved into line.

**Check:** the packet has four straight sides and corners at about right angles. No layer sticks out past the one under it. No elastic hem shows on the outside. The packet holds its shape with no gripper on it.

If a layer keeps springing out, the **right gripper** lifts it and lays it down once more. If it still springs out, open the last fold, stroke it flat, and fold it again.

### Step 8: End the episode

- Confirm the packet sits on the fold mat as a squared rectangle, the mat holds nothing else, and the rest of the tabletop is bare.
- Return both arms home with grippers open.
- Stop recording.

## After the episode: reset the workspace

This reset is not recorded.

1. Open the packet out flat.
2. Pull each corner out of the nest stack until all four corner pockets are free and open again.
3. Press the sheet flat again, so no fold line from the last episode is left in the cloth. A sheet that still shows an old fold is not staged again.
4. Check the elastic hem all the way round. Replace a sheet whose elastic is broken, pulled out of the seam, or so slack that a pocket mouth will not stand open when its hem is lifted.
5. Replace any sheet that is torn, stained, damp, has an open seam, or will not hold a fold.
6. Spread the sheet on the fold mat pockets up, square to the mat, with clear mat all round, no corner folded under, and all four pockets open.
7. Smooth the fold mat flat, check it has not shifted, and leave nothing else on it.
8. Wipe the tabletop and confirm the mat sits firm.
9. Run both Setup checklists again.

## SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

### How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

### Episode handling

Tag every violation with its timestamp and name. An episode may contain several violations; tag each one separately. Keep the episode with its violation tags. Do not delete it just because a rule was broken.

## Violations

### Violation: Nesting started on a sheet that is not spread flat

- **Visible cue:** the first gape or the first corner carry happens while the sheet is bunched, crooked on the mat, lying pockets down, has a corner folded under, or has part of it off a mat edge.
- **SOP rule broken:** Step 1, the sheet lies flat and square on the mat, pockets up, with all four corners showing and nothing off a mat edge, before any nest is made.
- **Coaching note:** straighten it first. A bad start makes every nest and every fold after it crooked.

### Violation: Nests done out of order

- **Visible cue:** the two stacks are joined before both pairs are nested; a near corner is carried to the far corner on the left before the right pair is done; or a corner is carried to a corner it is not paired with.
- **SOP rule broken:** Step 2, run 2.1, 2.2, and 2.3 in that order: the right pair, then the left pair, then the join.
- **Coaching note:** right pair, left pair, join. Finish one nest before you start the next.

### Violation: Corner tucked in without a gape

- **Visible cue:** the left gripper is off the receiving pocket while the right gripper carries a corner in; the gape is released before the right gripper is clear; or the left gripper pulls the hem up and lets it drop again instead of holding it open.
- **SOP rule broken:** Steps 2.1, 2.2, and 2.3, the left gripper holds the pocket mouth open for the whole substep and lowers it only after the right gripper has opened and come clear.
- **Coaching note:** open it, hold it, and let go last.

### Violation: Corner laid on the pocket instead of inside it

- **Visible cue:** a corner is set down on top of, beside, or under a pocket rather than slid into the pocket mouth, so its seam point stays visible from above, and the work moves on.
- **SOP rule broken:** Steps 2.1, 2.2, and 2.3, the corner is slid into the open pocket mouth until its seam point sits deep inside, and the outer pocket then covers it.
- **Coaching note:** all the way in, until you cannot see the point any more.

### Violation: Corner picked up by the wrong place

- **Visible cue:** a travelling corner or a nest stack is pinched at the cloth, at the hem, or at a side seam instead of at its seam point, and it hangs loose or pulls out of shape on the way over.
- **SOP rule broken:** Steps 2.1, 2.2, and 2.3, a corner is pinched at its seam point and nowhere else.

- **Coaching note:** find the point first, then lift.

#### **Violation: Nest left loose and the work moved on**

- **Visible cue:** the outer pocket is left rucked up, twisted, or sitting off to one side over the corner inside it, the stroke along its hem is skipped, and the next substep starts anyway.
- **SOP rule broken:** Steps 2.1, 2.2, and 2.3, the right gripper strokes along the outer pocket's hem so the pocket lies square over the corner inside it.
- **Coaching note:** one stroke on the hem settles the nest. Do not skip it.

#### **Violation: Fold or tuck made by dragging**

- **Visible cue:** a corner, a nest stack, a hem, or a panel is pushed, shoved, or slid across the mat into place instead of being pinched, lifted clear, carried over, and lowered onto its spot.
- **SOP rule broken:** Steps 2, 3, 5, and 6, every corner, hem, and panel is lifted clear of the cloth under it before it is carried.
- **Coaching note:** pick it up and put it down. Nothing gets pushed.

#### **Violation: Fold or tuck made without a pin**

- **Visible cue:** no gripper is pinning the sheet while a tuck bite or a thirds fold is made; the pin comes down only after the fold has started; or it lifts away before the working gripper is off the cloth, and the sheet shifts on the mat.
- **SOP rule broken:** Steps 3.1, 3.2, 5.1, 5.2, 6.1, and 6.2, one gripper pins the middle for the whole substep and lifts only after the other gripper is off the cloth.
- **Coaching note:** pin first, let go last.

#### **Violation: Edge tuck skipped or hem left showing**

- **Visible cue:** the flatten or a thirds fold starts while the far side or the left side still runs wavy, or a stretch of hem still bulges outside the outline; or a side gets fewer than three bites.
- **SOP rule broken:** Steps 3.1 and 3.2, both hem sides are tucked in three bites so the far side and the left side run straight, with no hem bulging outside the outline.
- **Coaching note:** three bites a side, then check the outline before you move on.

#### **Violation: Hem folded outward or a fold edge tucked**

- **Visible cue:** a hem is folded away from the sheet, or left hanging past the side; or a bite is taken on the near side or the right side, which are fold edges and carry no hem.
- **SOP rule broken:** Steps 3.1 and 3.2, the hem is folded inward onto the sheet, and only the far side and the left side are tucked.
- **Coaching note:** hem goes in, onto the sheet. The two fold edges are never tucked.

#### **Violation: Flatten skipped or dabbed**

- **Visible cue:** the first thirds fold starts with a bubble, ridge, or standing wrinkle still in the sheet; or the gripper taps the sheet in spots, presses once in the middle, or lifts off partway through a stroke instead of running each stroke from the middle out to a side.
- **SOP rule broken:** Step 4, four strokes, each one from beside the pin out to a side, in contact the whole way.
- **Coaching note:** middle outward, one stroke per side, no lifting off.

#### **Violation: Sheet knocked out of square**

- **Visible cue:** a stroke, a bite, or a carry pushes the quartered sheet out of shape, so a fold edge that ran straight now runs crooked or the near and right sides no longer meet at about a right angle, and the work moves on.
- **SOP rule broken:** Step 4, every stroke runs from the middle outward and never back toward the middle, and a corner that has moved is lifted clear, laid back in line, and the side stroked again.
- **Coaching note:** always stroke away from the middle. If it moves, lift and lay it back.

#### **Violation: Panels are not thirds**

- **Visible cue:** a folded edge lands well short of or well past two-thirds of the way across, or the second edge lands well off the first fold edge, so the three panels are clearly different widths, and the work moves on.
- **SOP rule broken:** Steps 5.1, 5.2, 6.1, and 6.2, the first fold lands its edge about two-thirds of the way across and the second lands level with the first fold edge.
- **Coaching note:** eye the two-thirds line before you lower it, then match the second fold to the first edge.

#### **Violation: Thirds folded in the wrong order**

- **Visible cue:** the right third is folded before the left third, or the near third before the far third, so the nest stack finishes on the outside of the packet instead of inside it.
- **SOP rule broken:** Steps 5 and 6, the left third goes in first and the far third goes down first, so the nest stack ends up in the middle of the packet.
- **Coaching note:** the lumpy corner goes inside. Left first, then far.

#### **Violation: Wrong number of folds or wrong axis**

- **Visible cue:** an axis is folded in half or in four instead of in thirds; the same axis is folded twice and the other axis is skipped; or the packet reaches Step 7 with only one axis folded.
- **SOP rule broken:** Steps 5 and 6, three panels across, then three panels front to back, one axis each.
- **Coaching note:** thirds across, thirds front to back. Two axes, three panels each.

#### **Violation: Fold edge not stroked**

- **Visible cue:** a gripper moves straight from lowering a panel to the next fold with no stroke along the new fold edge, or it taps the edge in spots instead of running one stroke end to end.
- **SOP rule broken:** Steps 5.1, 5.2, 6.1, and 6.2, the gripper that made the fold strokes along the new fold edge, end to end.
- **Coaching note:** every fold gets one stroke, end to end, before the next one.

#### **Violation: Squaring skipped**

- **Visible cue:** the episode ends with a layer standing out past the layer under it, a side running crooked, or elastic hem showing on the outside of the packet; or fewer than four sides get a stroke in Step 7.
- **SOP rule broken:** Step 7, all four sides are worked and stroked in order, far, right, near, left, so the packet reads as a rectangle with no layer sticking out.
- **Coaching note:** four sides, same order every time. Do not stop at the two that look wrong.

#### **Violation: Packet patted, nudged, or shoved**

- **Visible cue:** a gripper taps, brushes, pokes, or pushes a layer sideways to bring it into line, or straightens the packet after letting go of it, instead of pinching that layer, lifting it clear, and laying it back down.
- **SOP rule broken:** Steps 4 and 7, a layer that is out is lifted clear and laid back level with the layer under it.
- **Coaching note:** lift it and lay it again, or leave it alone.

#### **Violation: Cloth turned or regripped in the air**

- **Visible cue:** a corner, a nest stack, or a panel turns while a gripper holds it off the mat, or a gripper shifts its hold along the cloth without setting it down first.
- **SOP rule broken:** Steps 2, 3, 5, and 6, cloth is carried level and lowered as it is, and a new hold is taken only after it is resting.
- **Coaching note:** put it down before you turn it or take a new hold.

#### **Violation: Let go too high or over the wrong place**

- **Visible cue:** a gripper opens above the mat and the cloth drops the rest of the way; or a gripper opens over bare tabletop or over open space.
- **SOP rule broken:** Steps 1 to 7, a gripper opens over the fold mat only, and only once the cloth is resting.
- **Coaching note:** all the way down, then open.

#### **Violation: Arms swapped roles or crowded the same spot**

- **Visible cue:** the left gripper carries a travelling corner or a nest stack, tucks the far side, folds the right third, folds a front-to-back third, or squares the packet; the right gripper gapes a receiving pocket, tucks the left side, or folds the left third; or the two grippers work the same spot, touch each other, or one has to move before the other can reach.
- **SOP rule broken:** Steps 1 to 7, the right gripper carries every corner in, tucks the far side, strokes the sheet flat, folds the right third and both front-to-back thirds, and squares the packet, and the left gripper gapes, pins, and works the left side only.
- **Coaching note:** right works, left holds. The left arm only folds on the left.

#### **Violation: Cloth dropped or the mat knocked out of place**

- **Visible cue:** the sheet or the packet falls off the mat or onto the floor; a corner slips out of a gripper mid-carry; a gripper strikes the fold mat and it shifts or bunches up; or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 7, the cloth stays on a clear travel path on the mat, and a gripper opens only over cloth that is already resting.
- **Coaching note:** check the path and the landing spot before you move.

#### **Violation: Check skipped and a fault carried forward**

- **Visible cue:** a step's check is not looked at and the next step starts over a visible fault: a corner outside its pocket, hem bulging outside the outline, a bubble still in the sheet, or a panel hanging out past the layer under it.
- **SOP rule broken:** Steps 1 to 7, each step's check is confirmed and its retry is run before the next step starts.
- **Coaching note:** look at the check every time. A fault only gets harder to fix once it is folded over.

#### **Violation: Wrong episode ending**

- **Visible cue:** the episode ends with the packet unsquared, a layer still standing out, something other than the packet on the mat, either arm away from home, a gripper closed, or an arm doing something else after homing.
- **SOP rule broken:** Step 8, confirm the squared packet on the bare mat, return both arms home with grippers open, then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

#### **Non-violation failures**

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).

- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective sheet:** found torn, stained, damp, or with an open seam.
- **Broken or slack elastic:** the elastic is pulled out of the seam, or a pocket mouth will not stand open when the left gripper lifts its hem. Replace the sheet before the next episode.
- **Sheet will not hold a fold:** stroked end to end and the fold still springs open every time. Press it before the next episode.
- **Sheet too big for the mat:** it cannot lie spread inside the mat with clear mat all round.
- **Fold mat shifts out of place** or bunches up under the sheet during the episode.

## Annotation subtasks (from SOP)

1. Straighten a corner of the spread sheet
2. Hold a corner pocket open
3. Carry one corner into another corner's pocket
4. Stroke a nested pocket square
5. Tuck the elastic hem in along one side
6. Stroke the sheet flat
7. Fold one third and stroke the fold edge
8. Square the finished packet
9. Return both arms home and end the episode